

ABERDARE TECHNICAL COLLEGE

MODULE 3,TERM 3

UNIT:NETWORK DESIGN AND MANAGEMENT

CONTINUOUS ASSESSMENT TEST-CAT2

UNIT CODE: 0612 451 07A

ELEMENT:INSTALL COMPUTER NETWORK

Total marks: 50

Time 1hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. This paper has **fifteen questions**. Answer **ALL** questions.
2. Write your name, registration number, class and date on the answer booklet.
3. The marks for each question or part are shown in brackets. **Each mark matches one point, item or command asked for**. For example, "state four" carries 4 marks.
4. In calculation questions, **write the formula first, then substitute, then give the answer**.
An answer without a formula or working scores at most half of the marks for that part.
5. IP addresses and subnet masks must be written in **dotted-decimal notation** unless stated otherwise.
6. Router and switch commands must be written in **Cisco IOS syntax**.
7. Use the marks to decide how much to write and how long to spend (about 2 minutes per mark).
8. Calculators are allowed. Mobile phones are **not** allowed in the examination room.
9. Write neatly. Untidy work may lose marks.

QUESTIONS

1. List the **four** layers of the TCP/IP model. (4 marks)
2. State **two** functions of the Network layer of the OSI model. (2 marks)
3. Define each of the following terms:
 - (i) Classful IP addressing (1 mark)
 - (ii) Classless Inter-Domain Routing (CIDR) (1 mark)
4. State **two** differences between static and dynamic IP addressing. (2 marks)
5. State **two** differences between IPv4 and IPv6. (2 marks)
6. (a) State the default subnet mask of each of the following address classes: Class A, Class B and Class C. (3 marks)
(b) Identify the class of the address **192.168.5.7**. (1 mark)
7. (a) Write the formula used to calculate each of the following in CIDR:
 - (i) Number of host bits (1 mark)
 - (ii) Total number of host addresses in a network (1 mark)
 - (iii) Number of usable host addresses in a network (1 mark)
(b) A network has the address **192.168.10.0/26**. Showing your working, determine:
 - (i) The number of host bits (1 mark)
 - (ii) The total number of host addresses (1 mark)
 - (iii) The number of usable host addresses (1 mark)
8. Convert each of the following CIDR prefix lengths to a subnet mask in dotted-decimal notation:
 - (a) /24 (1 mark)
 - (b) /26 (1 mark)
 - (c) /28 (1 mark)
 - (d) /30 (1 mark)

9. Convert each of the following subnet masks to a CIDR prefix length:

- (a) 255.255.255.0 (1 mark)
- (b) 255.255.255.224 (1 mark)
- (c) 255.255.255.252 (1 mark)

10. A network must support **at least 50 usable host addresses**. Showing your working, determine:

- (a) The number of host bits needed. (1 mark)
- (b) The CIDR prefix length. (1 mark)
- (c) The subnet mask in dotted-decimal notation. (1 mark)
- (d) The number of usable host addresses the network provides. (1 mark)

11. State **two** advantages of CIDR over classful IP addressing. (2 marks)

12. (a) Name the protocol used to assign IP addresses to devices dynamically. (1 mark)

(b) State **one** reason why a router interface should use a static IP address. (1 mark)

13. Write the Cisco IOS command for each of the following tasks on a router:

- (a) Set the hostname to **R1**. (1 mark)
- (b) Set an encrypted privileged EXEC password **kcse2026**. (1 mark)
- (c) Enter configuration mode for interface **GigabitEthernet0/0**. (1 mark)
- (d) Assign the IP address **192.168.10.1** with the mask for a /26 network. (1 mark)
- (e) Activate the interface. (1 mark)
- (f) Save the running configuration to NVRAM. (1 mark)

14. Write the **three** commands used to configure the management interface (VLAN 1) of a switch with the IP address **192.168.10.2** and the mask for a /26 network, and make the interface active. (3 marks)

15. Router R1 connects to the network **10.0.0.0/8** and to the LAN **192.168.10.0/24**. Write the **four** commands needed to configure **RIP version 2** on R1 so that both networks are advertised. (4 marks)

